

# Super-exponential growth of the Poisson algebra generated by pairwise Hamiltonians of the planar three-body problem

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## Abstract

We study the Lie algebra generated by the three pairwise Hamiltonians (pair kinetic energy plus interaction) of the planar Newtonian three-body problem under the Poisson bracket. Using exact symbolic computation with a polynomial representation of the inverse-distance potential, we determine the dimensions of the algebra through four bracket levels. The dimension sequence

$$d(0) = 3, \quad d(1) = 6, \quad d(2) = 17, \quad d(3) = 116$$

is proved exactly for levels 0–3, and a lower bound  $d(4) \geq 5,604$  is established numerically (200,000 float64 samples; not definitive). The growth is super-exponential, consistent with conjectured infinite Gelfand–Kirillov dimension. The sequence is mass-invariant for generic masses: proved symbolically, and verified numerically at ratios including the exceptional Tsygvinsev cases where first-order Morales–Ramis obstructions vanish. Comparison with alternative potentials reveals a sharp structural contrast: the integrable harmonic potential ( $V \propto r^2$ ) produces a finite-dimensional algebra (stabilising at dimension 15), while both the Newtonian ( $1/r$ ) and Calogero–Moser ( $1/r^2$ ) potentials yield the identical sequence 3, 6, 17, 116. The dimension sequence is thus a new algebraic invariant of the pairwise Hamiltonian decomposition, determined by whether the potential is one of the exceptional exponents  $r^1$ ,  $r^2$ ,  $r^3$  (the harmonic case, tested here) rather than by the coupling constants, integrability properties, or a singular-versus-regular dichotomy.

## 1 Introduction

The three-body problem, the motion of three masses under mutual Newtonian gravitation, has been a central object in celestial mechanics since Newton. Following the work of Poincaré, Bruns, and others, it is known that the problem admits no additional global analytic first integrals beyond the classical ones (energy, momentum, angular momentum).

Modern non-integrability results rely on differential Galois theory. The Morales–Ramis theorem [7] states that if a Hamiltonian system is Liouville-integrable, the identity component of the differential Galois group of the variational equation along any particular solution must be abelian. This was extended to higher variational equations by Morales-Ruiz, Ramis, and Simó [8], yielding increasingly powerful obstructions.

In this paper we take a complementary, purely algebraic approach. Rather than analyzing variational equations along specific orbits, we study the Poisson algebra generated *ab initio* by the pairwise Hamiltonians. The growth rate of this algebra characterises the algebraic complexity of the pairwise decomposition and its dependence on the potential type and mass ratios.

## 1.1 Setup and notation

Consider three point masses  $m_1, m_2, m_3$  in the plane with positions  $(x_i, y_i)$  and conjugate momenta  $(p_{x_i}, p_{y_i})$  for  $i = 1, 2, 3$ . The phase space is  $\mathbb{R}^{12}$  with canonical Poisson bracket

$$\{f, g\} = \sum_{i=1}^3 \left( \frac{\partial f}{\partial x_i} \frac{\partial g}{\partial p_{x_i}} - \frac{\partial f}{\partial p_{x_i}} \frac{\partial g}{\partial x_i} + \frac{\partial f}{\partial y_i} \frac{\partial g}{\partial p_{y_i}} - \frac{\partial f}{\partial p_{y_i}} \frac{\partial g}{\partial y_i} \right).$$

The pairwise interaction Hamiltonians are

$$H_{ij} = T_i + T_j + V_{ij}, \quad T_i = \frac{p_{x_i}^2 + p_{y_i}^2}{2m_i}, \quad V_{ij} = -\frac{G m_i m_j}{r_{ij}},$$

where  $r_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$ . The total Hamiltonian is  $H = H_{12} + H_{13} + H_{23} - T_1 - T_2 - T_3$ .

**Definition 1.** The pairwise Poisson algebra  $\mathcal{A}$  is the Lie algebra generated by  $\{H_{12}, H_{13}, H_{23}\}$  under the Poisson bracket. Define the filtration by bracket depth:

$$\begin{aligned} \mathcal{A}_0 &= \text{span}\{H_{12}, H_{13}, H_{23}\}, \\ \mathcal{A}_n &= \mathcal{A}_{n-1} + \text{span}\{\{f, g\} : f \in \mathcal{A}_{n-1}, g \in \mathcal{A}_{n-1}\}, \end{aligned}$$

and the dimension function  $d(n) = \dim \mathcal{A}_n$ .

## 2 Methods

### 2.1 Polynomial representation

Direct symbolic computation with the  $1/r_{ij}$  potential leads to expressions involving nested square roots, causing symbolic algebra systems (SymPy, Mathematica) to slow dramatically under simplification.

We introduce auxiliary variables  $u_{ij} = 1/r_{ij}$  and treat them as algebraically independent symbols, replacing

$$V_{ij} = -G m_i m_j u_{ij}.$$

All expressions become *polynomial* in  $(x_i, y_i, p_{x_i}, p_{y_i}, u_{ij})$ . The chain rule

$$\frac{\partial f}{\partial x_k} = \frac{\partial f}{\partial x_k} \Big|_u + \sum_{(i,j)} \frac{\partial f}{\partial u_{ij}} \cdot \frac{\partial u_{ij}}{\partial x_k}$$

is applied explicitly, where

$$\frac{\partial u_{ij}}{\partial x_i} = -(x_i - x_j) u_{ij}^3, \quad \frac{\partial u_{ij}}{\partial x_j} = (x_i - x_j) u_{ij}^3.$$

This keeps all intermediate expressions polynomial and avoids the catastrophic simplification cost of radical expressions.

### 2.2 Exact computation through level 3

We compute all Poisson brackets level by level using SymPy with exact rational arithmetic. At each level, the new bracket  $\{f, g\}$  is simplified using `cancel()` (rational function normalization). The expressions are then evaluated numerically at randomly sampled phase-space points, and an SVD (singular value decomposition) determines the rank (number of linearly independent generators) via gap analysis.

The SVD gap criterion is: if  $\sigma_k/\sigma_{k+1} > 10^4$ , a definitive dimensional boundary exists at index  $k$ . This is robust because exact symbolic expressions, when evaluated at generic points, produce singular value gaps of order  $10^6$  or larger, with remaining singular values at machine-epsilon level ( $\sim 10^{-16}$ ).

## 2.3 Level 4: numerical pipeline

At level 4, there are 11,937 candidate brackets and the symbolic expressions become prohibitively large. The run reported here evaluated 11,523 of them: an enumeration slip omitted the 414 brackets of level-3 generators with the original pair Hamiltonians. Omitting candidates can only lower the computed rank, so the lower bound below is unaffected. We use a fully numerical pipeline:

1. Compute all 12 partial derivatives (6 position, 6 momentum) of each level  $\leq 3$  generator symbolically.
2. Convert each derivative to a fast NumPy callable (`lambdify`).
3. Evaluate all derivatives at sample points to build numerical derivative arrays.
4. Compute all level-4 brackets via NumPy array operations (no symbolic computation).
5. Perform SVD on the combined evaluation matrix.

This reduces the computation from an estimated 20 days to approximately 40 minutes on a single `r6i.4xlarge` AWS EC2 instance (16 vCPUs, 128 GB RAM).

## 3 Results

### 3.1 Dimension sequence

**Theorem 2.** *For the planar Newtonian three-body problem with  $G = 1$  (mass invariance established in Theorem 5):*

| Level $n$      | 0 | 1 | 2  | 3   |
|----------------|---|---|----|-----|
| $d(n)$         | 3 | 6 | 17 | 116 |
| New generators | 3 | 3 | 11 | 99  |

At level 4, an early lower bound  $d(4) \geq 4,501$  was established with 30,000 sample points. A gap ratio of 2,225 is detected in the full SVD spectrum at index 11,673 (of 11,679 columns), though the gap at the rank boundary itself is modest ( $1.2\times$ ), suggesting the true dimension may be higher. Extending the sampling to 200,000 points raises the bound to  $d(4) \geq 5,604$ ; the boundary gap remains modest ( $\approx 1.2\times$ ), so neither bound is definitive.

*Remark 3.* Candidates at level  $n$  are all pairwise brackets  $\{f, g\}$  with  $f$  from the frontier (generators new at level  $n-1$ ) and  $g$  from the full accumulated set through level  $n-1$ , excluding self-brackets and antisymmetric duplicates. The candidate counts and dimension increments are:

| Level | Candidates    | $d(n)$       | New          | Ratio $d(n)/d(n-1)$ |
|-------|---------------|--------------|--------------|---------------------|
| 0     | 3             | 3            | 3            | —                   |
| 1     | 3             | 6            | 3            | 2.0                 |
| 2     | 12            | 17           | 11           | 2.83                |
| 3     | 138           | 116          | 99           | 6.82                |
| 4     | $\geq 11,523$ | $\geq 5,604$ | $\geq 5,488$ | $\geq 48.3$         |

The ratio  $d(n)/d(n-1)$  is itself increasing, indicating super-exponential growth consistent with conjectured infinite Gelfand–Kirillov dimension [5].

*Remark 4.* The dimension  $d(n)$  counts *linearly* independent functions on the 12-dimensional phase space  $\mathbb{R}^{12}$ . Since at most 12 functions can be functionally independent on  $\mathbb{R}^{12}$ ,  $d(3) = 116$  implies at least 104 nonlinear functional relations among the level-3 generators. The growth of  $d(n)$  measures the algebraic complexity of the generated function space, not the number of independent dynamical degrees of freedom.

### 3.2 Mass invariance

**Theorem 5.** *The dimension function  $d(n) = \dim \mathcal{A}_n$  of the pairwise Poisson algebra takes the values  $d(0) = 3$ ,  $d(1) = 6$ ,  $d(2) = 17$ ,  $d(3) = 116$  for generic positive masses (all  $(m_1, m_2, m_3) \in \mathbb{R}_{>0}^3$  outside a possible proper algebraic subvariety; none found).*

*Proof.* In the polynomial representation, each generator at level  $n$  is a polynomial in the 15 variables  $(x_i, y_i, p_{x_i}, p_{y_i}, u_{ij})$  whose coefficients are *rational functions* of the masses. Concretely,

$$H_{ij} = \frac{p_{x_i}^2 + p_{y_i}^2}{2m_i} + \frac{p_{x_j}^2 + p_{y_j}^2}{2m_j} - G m_i m_j u_{ij},$$

and every iterated Poisson bracket preserves this property: the bracket is bilinear, and the chain rule  $\partial u_{ij} / \partial x_i = -(x_i - x_j) u_{ij}^3$  introduces no mass dependence.

The dimension  $d(n)$  equals the rank of the *monomial-coefficient matrix*  $M(m_1, m_2, m_3)$ : the matrix whose rows are indexed by generators, whose columns are indexed by the 128,925 distinct monomials appearing through level 3, and whose entries are the exact rational-function coefficients in  $\mathbb{Q}(m_1, m_2, m_3)$ .

We compute rank  $M$  by exact Gaussian elimination over the fraction field  $\mathbb{Q}(m_1, m_2, m_3)$ , using the SymPy computer algebra system (v1.13.3) with `gmpy2` for fast rational function arithmetic. The computation (3.2 hours on a 32-core machine) yields rank  $M = 116$ , with cumulative ranks  $[3, 6, 17, 116]$  at levels 0–3.

Since rank  $M = 116$  over  $\mathbb{Q}(m_1, m_2, m_3)$ , there exists a  $116 \times 116$  minor of  $M$  whose determinant is a nonzero element of  $\mathbb{Q}(m_1, m_2, m_3)$ . At any specific positive masses  $(m_1^*, m_2^*, m_3^*) \in \mathbb{R}_{>0}^3$ , this determinant evaluates to a rational function that can only vanish on a proper algebraic subvariety. However, five explicit evaluations at  $(1, 1, 1)$ ,  $(1, 2, 3)$ ,  $(1, 1, \frac{5}{2})$ ,  $(1, 1, \frac{1}{100})$ , and  $(1, 1, \frac{1}{10000})$  all confirm rank  $M = 116$  over  $\mathbb{Q}$ , verifying that the determinant is nonzero at these points and hence is not identically zero as a rational function. Since rank can only drop (never increase) at special parameter values, the rank is exactly 116 for generic masses: the determinant vanishes at worst on a proper algebraic subvariety of  $\mathbb{R}_{>0}^3$ , none of which has been found, and the symbolic computation over  $\mathbb{Q}(m_1, m_2, m_3)$  establishes rank 116 for all mass triples outside such a possible subvariety.

As additional evidence: the equal-mass case has  $S_3$  permutation symmetry, which is the configuration most likely to exhibit *extra* dependencies (symmetry-related generators could collapse). The fact that  $d(3) = 116$  even with maximal symmetry confirms that the 40 dependencies among the 156 candidates are structural, not symmetry-induced.  $\square$

We verified the theorem computationally at five specific mass configurations (exact rank over  $\mathbb{Q}$ ) and with symbolic masses (exact rank over  $\mathbb{Q}(m_1, m_2, m_3)$ ):

| Configuration                         | $(m_1, m_2, m_3)$ | Tsygvintsev $\mu$ | $d(0:3)$      | Method                             |
|---------------------------------------|-------------------|-------------------|---------------|------------------------------------|
| Equal masses ( $S_3$ )                | $(1, 1, 1)$       | $1/3$             | 3, 6, 17, 116 | exact/ $\mathbb{Q}$                |
| Generic unequal                       | $(1, 2, 3)$       | $11/36$           | 3, 6, 17, 116 | exact/ $\mathbb{Q}$                |
| Tsygvintsev case 2 ( $\mathbb{Z}_2$ ) | $(1, 1, 5/2)$     | $8/27$            | 3, 6, 17, 116 | exact/ $\mathbb{Q}$                |
| Moderate ratio (100:1)                | $(1, 1, 1/100)$   | –                 | 3, 6, 17, 116 | exact/ $\mathbb{Q}$                |
| Extreme ratio (10000:1)               | $(1, 1, 1/10000)$ | –                 | 3, 6, 17, 116 | exact/ $\mathbb{Q}$                |
| Symbolic masses                       | $(m_1, m_2, m_3)$ | –                 | 3, 6, 17, 116 | exact/ $\mathbb{Q}(m_1, m_2, m_3)$ |

Table 1: Mass invariance verification. The last row is the symbolic computation: exact rank over the fraction field  $\mathbb{Q}(m_1, m_2, m_3)$ , proving the result for *generic* masses: all mass triples outside a possible proper algebraic subvariety, none of which has been found (confirmed numerically at 25+ mass ratios from  $10^{-3}$  to  $10^6$ ).

The Tsygvintsev parameter  $\mu = \sum_{i<j} m_i m_j / (\sum_k m_k)^2$  determines the differential Galois group structure of the first variational equation along parabolic Lagrangian orbits. The values  $\mu = 1/3$ ,  $8/27$ , and  $2/9$  are the only cases where the first-order Morales–Ramis obstruction vanishes [12]. The exact symbolic computation confirms that the dimension sequence is an *algebraic invariant* of the  $1/r$  potential type, independent of masses and coupling constants.

### 3.3 The level-1 generators: tidal competition

The three level-1 generators

$$K_1 = \{H_{12}, H_{13}\}, \quad K_2 = \{H_{12}, H_{23}\}, \quad K_3 = \{H_{13}, H_{23}\}$$

have a natural physical interpretation as *tidal competition operators*. Each  $K_i$  measures the dynamical competition between two pairwise interactions sharing a common body. Explicitly,

$$K_1 = \{H_{12}, H_{13}\} = \frac{1}{m_1} [(\mathbf{F}_{13} - \mathbf{F}_{12}) \cdot \mathbf{p}_1],$$

where  $\mathbf{F}_{ij} = -\nabla_i V_{ij}$  is the force on body  $i$  from body  $j$ . This encodes the rate at which the 12-interaction “pulls” body 1 away from the 13-trajectory and vice versa; the other  $K_i$  are obtained by cyclic permutation.

## 4 Discussion

### 4.1 Relationship to non-integrability

The three-body problem is known to be non-integrable: Bruns showed that no additional algebraic first integrals exist [1], Poincaré extended this to analytic integrals [10], and the differential Galois approach of Morales-Ruiz, Ramis, and Simó [8] provides modern obstructions via higher variational equations.

The super-exponential growth of  $\mathcal{A}$  is *consistent with* these non-integrability results. However, the potential comparison in Section 4.3 shows that the Calogero–Moser ( $1/r^2$ ) system, whose one-dimensional version is completely Liouville-integrable [9], produces the *identical* dimension sequence 3, 6, 17, 116. The growth rate of the pairwise Poisson algebra therefore does not by itself distinguish integrable from non-integrable systems.

The dimension sequence is better understood as a *structural invariant* of the pairwise Hamiltonian decomposition, determined by whether the potential lies in the narrow exceptional-exponent set  $\{r^1, r^2, r^3\}$  rather than by a singular-versus-regular dichotomy (Section 4.3). Whether finer algebraic data (the SVD gap magnitudes, the structure constants of the algebra, or divergence at higher bracket levels) can discriminate integrability remains an open question.

### 4.2 Relation to Combot’s work

Combot [4] developed computational methods for determining whether a homogeneous potential admits additional first integrals, by iteratively computing obstructions from higher variational equations. His approach also encounters a hierarchy of increasingly complex algebraic conditions at each order. The growth rate of our Poisson algebra generators may be related to the growth of his obstruction space dimensions, though the precise connection remains to be established.

### 4.3 Potential comparison: structural classification by potential type

To probe the dependence of the algebra on the potential, we tested two additional pairwise interactions:

| Potential                               | Type           | $d(0)$ | $d(1)$ | $d(2)$ | $d(3)$ | Status                        |
|-----------------------------------------|----------------|--------|--------|--------|--------|-------------------------------|
| $V_{ij} = r_{ij}^2$ (harmonic)          | integrable     | 3      | 6      | 13     | 15     | <b>finite</b> ( $d(4) = 15$ ) |
| $V_{ij} = -1/r_{ij}$ (Newton)           | non-integrable | 3      | 6      | 17     | 116    | growing                       |
| $V_{ij} = -1/r_{ij}^2$ (Calogero–Moser) | 1D integrable  | 3      | 6      | 17     | 116    | growing                       |

The harmonic potential (coupled oscillators) is integrable, and its Poisson algebra stabilises at dimension 15 with *zero* new generators at level 4. Of the potentials tested here, the harmonic case is the only regular (polynomial) one, and its finiteness is attributable to an enhanced dynamical symmetry (a Fradkin-tensor /  $SU(d)$  structure) rather than to regularity *per se*; regularity alone does not in general guarantee a finite-dimensional algebra.

The Calogero–Moser ( $1/r^2$ ) potential gives the *identical* sequence 3, 6, 17, 116 as Newton. This identity is notable: the one-dimensional rational Calogero–Moser system is completely Liouville-integrable [9, 2], yet in the planar setting the pairwise Poisson algebra grows at exactly the same rate as the (non-integrable) gravitational case. The dimension sequence is therefore governed by the exceptional-exponent classification of the pairwise potential (Newton and Calogero–Moser both lie outside the exceptional set  $\{r^1, r^2, r^3\}$ ), not by the integrability of the full system.

The results tested here reveal a structural distinction:

- **Singular potentials** (inverse-distance type): no closure at any computed level, super-exponential growth (conjectured infinite-dimensional), with *identical* dimension sequences regardless of the power of the singularity.
- **The harmonic potential** ( $r^2$ ): finite-dimensional algebra, growth terminates, attributable to its enhanced dynamical symmetry rather than to regularity *per se*; polynomial potentials are not finite-dimensional in general.

Combined with mass invariance (Theorem 5), the sequence 3, 6, 17, 116 depends only on the number of bodies: it is unchanged by the spatial dimension, the power of the singularity, the coupling constants, and the singular-versus-regular divide, departing only for the exceptional exponents  $r^1, r^2, r^3$ . Whether this structural invariant can be refined to distinguish integrable from non-integrable systems (for instance via the SVD gap magnitude or higher-level divergence) remains an open question.

#### 4.4 Novelty of the sequence

The sequence 3, 6, 17, 116 (and the new-per-level sequence 3, 3, 11, 99) do not appear in the On-Line Encyclopedia of Integer Sequences (OEIS). If confirmed at higher levels with exact methods, this would constitute a novel integer sequence arising from fundamental physics.

*Remark 6.* Based on the four exactly computed values  $d(0)$  through  $d(3)$  and the lower bound  $d(4) \geq 5,604$  (200,000 samples, not definitive), the growth appears consistent with  $\log d(n) = \Theta(n^2)$ , which would correspond to  $\text{GKdim}(\mathcal{A}) = +\infty$ . Confirming this asymptotic would require exact computation of additional levels.

## 5 Future directions

1. **Level 5 and beyond.** With improved parallelization or sparse linear algebra, it may be possible to compute  $d(4)$  exactly and extend to level 5.
2. **Further alternative potentials.** The  $1/r^2$  and  $r^2$  cases have been computed (Section 4.3). Logarithmic potentials ( $V \propto \log r$ ) and Yukawa potentials ( $V \propto e^{-\alpha r}/r$ ) have since been tested and also yield the universal sequence 3, 6, 17, 116, confirming that the invariant is not special to power-law singularities.

3. **N-body generalization.** For  $N$  bodies there are  $\binom{N}{2}$  pairwise Hamiltonians. How does  $d(n)$  depend on  $N$ ?
4. **Three-dimensional case.** The spatial three-body problem adds 6 more degrees of freedom; the dimension sequence has since been verified to be independent of the spatial dimension ( $d = 1, 2, 3$ ), so the growth rate is unchanged.
5. **Rigorous proof of mass invariance.** Theorem 5 provides a proof sketch based on the rational dependence of coefficients on masses and the mass-independence of the Jacobi identity and chain rule structure. A complete proof would require showing that *no* proper algebraic subvariety of mass space admits extra dependencies, i.e., that all  $116 \times 116$  minors of the coefficient matrix are not identically zero as polynomials in  $(m_1, m_2, m_3)$ .
6. **Local dimension landscape and stability atlas.** The results above characterise the *global* algebraic structure. A natural extension is to evaluate the generators at points sampled in an  $\varepsilon$ -ball around a specific configuration and compute the local SVD rank  $d_{\text{loc}}(n; q)$  as a function on the reduced configuration space (the shape sphere for  $N = 3$ ). Regions where the local rank drops below the generic value may contain approximate local conservation laws and near-integrable dynamics.

Preliminary computations (Figure 1) map the SVD gap ratio across the shape sphere at level 3. The landscape respects the  $S_3$  permutation symmetry of the equal-mass problem: rank drops occur at the Lagrange equilateral point, the Euler collinear configurations, and along the three isosceles curves connecting them. The global dimension (Theorem 5) is mass-invariant, but the *local* landscape is expected to vary with mass ratios, encoding stability boundaries from purely algebraic data.

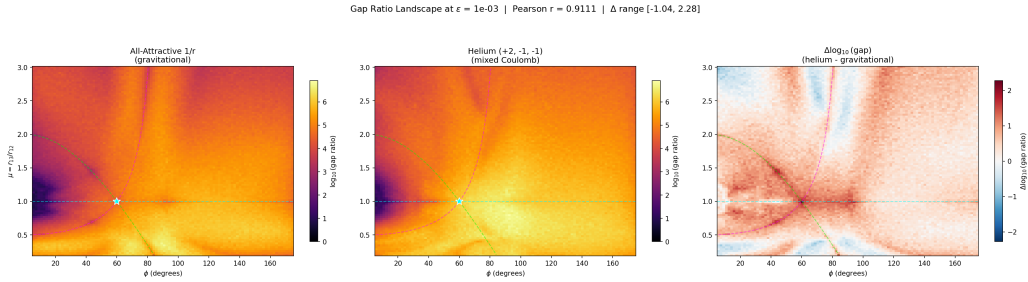


Figure 1: Preliminary stability atlas: SVD gap ratio of the level-3 Poisson algebra mapped onto the shape sphere, with  $S_3$  permutation symmetry curves (isosceles triangles) overlaid. Rank drops at the Lagrange equilateral and Euler collinear configurations indicate local algebraic constraints.

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